

BUSFIN 8200 - Problem Set 1

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Problem 1

1 (a)

Starting from $R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$ and assuming that D is greater than zero, we take log of both sides and have:

$$\begin{aligned} r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - \log(P_t) \\ &= \log(P_{t+1}) \left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - p_t \\ &= p_{t+1} + \log\left(1 + \frac{e^{d_{t+1}}}{e^{p_{t+1}}}\right) - p_t \\ &= p_{t+1} + \log(1 + e^{d_{t+1} - p_{t+1}}) - p_t \\ &= p_{t+1} + \log(1 + e^{dp_{t+1}}) - p_t \\ &= p_{t+1} - p_t + f(dp_{t+1}) \end{aligned}$$

where $f(dp_{t+1})$ is a log function of the log dividend-price ratio. Applying a first-order Taylor expansion around $\bar{dp}$ to $f(dp_{t+1})$, we have Equation (1):

$$r_{e,t+1} \approx p_{t+1} - p_t + f(\bar{dp}) + f'(\bar{dp})(dp_{t+1} - \bar{dp})$$

We define $\kappa = \frac{1}{1 + e^{\bar{dp}}}$. Consequently, we have the following:

$$\begin{aligned} f'(\bar{dp}) &= \frac{e^{\bar{dp}}}{1 + e^{\bar{dp}}} = 1 - \kappa \\ e^{\bar{dp}} &= \frac{1}{\kappa} - 1 \\ \bar{dp} &= \log\left(\frac{1}{\kappa} - 1\right) \end{aligned}$$

Plugging in the above results into Equation (1), we have:

$$\begin{aligned} r_{e,t+1} &\approx p_{t+1} - p_t + \log(1 + e^{\bar{dp}}) + (1 - \kappa)dp_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= p_{t+1} - p_t + \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)p_{t+1} + (1 - \kappa)d_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= \kappa p_{t+1} - p_t + (1 - \kappa)d_{t+1} + \kappa_0 \\ &= \kappa_0 - \kappa dp_{t+1} + (d_{t+1} - d_t) + (d_t - p_t) \\ &= \kappa_0 - \kappa dp_{t+1} + \Delta d_{t+1} + dp_t \end{aligned}$$

where $\kappa_0 = \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) = -\log(\kappa) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right)$. Rearranging the above equation, we have:

$$\begin{aligned} dp_t &= -\kappa_0 + \kappa dp_{t+1} + r_{e,t+1} - \Delta d_{t+1} \\ \Rightarrow dp_{t+1} &= -\kappa_0 + \kappa dp_{t+2} + r_{e,t+2} - \Delta d_{t+2} \\ \Rightarrow dp_{t+2} &= -\kappa_0 + \kappa dp_{t+3} + r_{e,t+3} - \Delta d_{t+3} \end{aligned}$$

Keeping substituting these equations into the first one and generalizing the pattern H times, we have:

$$\begin{aligned}
dp_t &= -\kappa_0 \underbrace{(1 + \kappa + \kappa^2)}_{\sum_{h=1}^H \kappa^{h-1}} + \kappa^3 dp_{t+3} + [r_{e,t+1} + \kappa r_{e,t+2} + \kappa^2 r_{e,t+3}] - [\Delta d_{t+1} + \kappa \Delta d_{t+2} + \kappa^2 \Delta d_{t+3}] \\
&= -\kappa_0 \left(\frac{1 - \kappa^H}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \\
&= \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \tag{1.1}
\end{aligned}$$

Given that $\mathbb{E}_t[dp_t] = dp_t$, we take conditional expectation on both sides of Equation (1.1) and have:

$$dp_t = \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.2}$$

Since $e^{\overline{dp}}$ is always positive, we have $0 < \kappa < 1$. Therefore, as $H \rightarrow \infty$, $\kappa^H \rightarrow 0$. In addition, we also need to impose No-Bubble Condition such that $\lim_{H \rightarrow \infty} \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] = 0$. Giglio, Maggiori, and Stroebel (2016) indeed shows that there were no systemic deviations in the discounts of extremely long leaseholds to otherwise identical freeholds in the U.K. and Singapore housing markets, thus indicating no rational bubbles. Consequently, letting $H \rightarrow \infty$, Equation (1.2) can be simplified as follows:

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.3}$$

1 (b)